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Transmission of indigenous knowledge systems under changing landscapes within the vhavenda community, South Africa

Olgah Lerato Malapane *, Nelson Chanza, Walter Musakwa

Department of Geography and Environmental Management and Energy Studies, Faculty of Science, University of Johannesburg, Auckland Park, Johannesburg 2093, South Africa

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ABSTRACT

Indigenous Knowledge (IK) and techniques benefit not only humankind but also benefit the environment. Nevertheless, the transmissions of indigenous knowledge are threatened and replaced by Western knowledge and ideologies. The focus of the study was to demonstrate the various Indigenous knowledge transmission methods, the driving forces behind their disappearance, and how they added value to the resilience of such knowledge forms in sustaining Indigenous and local communities throughout the centuries. To achieve this, the study used questionnaires consisting of open-ended and closed-ended questions (n = 136). Participants were selected and identified through purposeful sampling from five remote indigenous villages in South Africa. The data were collected only from principal elderly participants above 50 due to their profound knowledge and experiences with their local environment. The most prevalent indigenous knowledge within the Vhavenda people is folklore, which is orally transmitted from one generation to the next through traditional songs, stories, dances, myths, customs, and rituals. Drivers such as invasion of technology, reluctance on the part of custodians to pass on the knowledge, knowledge being forgotten, and knowledge being less effective were ranked to be highest in the hindrance of IK transmission in the study area. Solutions such as revitalising knowledge systems through proper documentation, such as building online libraries, integrating Western and indigenous knowledge, and introducing indigenous knowledge into mainstream education systems and media could assist in preserving such knowledge.

1. Introduction

Indigenous Knowledge (IK) is locally developed over centuries and has been transmitted orally from generation to generation. Between 50 and 90 percent of the 7000 indigenous languages spoken worldwide and practised are predicted to vanish by 2115 (United Nations, 2019; Trainor, 2020). Most Indigenous Knowledge (IK) languages and methods preserve distinctive customs and environmental information that would be lost if not transmitted from one generation to another (Havemann, 2016). IK is a set of unique knowledge that belongs to a specific ethnic group, culture, or society (Boven and Morohashi, 2002; Sraku-Lartey et al., 2017). IK is primarily relational, interconnected with language and land, according to Greenwood and Lindsay (2019), and intergenerationally transmitted orally through ceremonies, customs, songs, and ways of life (Owusu-Ansah and Mji, 2013). IK embodies several ways, cultures, and beliefs that shape this knowledge form (Indigenous Corporate Training Inc., 2018). The concept is recognised globally by prominent institutions such as the Intergovernmental Panel on Biodiversity and Ecosystem Services (IPBES), the Intergovernmental Panel on Climate Change (IPCC), and the World Bank. It is acknowledged as a valuable knowledge system resource that is significant in the sustainability of rural and disadvantaged communities, managing climate change and human development, respectively (Andrae-Marobela et al., 2012; Kaya and Seleti, 2013b). IK must be thoroughly understood as a logically valid living process (Battiste, 2002; Bates, 2009). Knowledge is derived from epistemology, which refers to the theory of knowledge, particularly the methods, validity, scope, and distinction between justified belief and opinion (Steup and Neta, 2020). IK refers to the knowledge local people acquire through experience, informal experiments, and an intimate understanding of their environment (Ogen, 2006; Mistry, 2009; Gope et al., 2017). Our study focuses on the Vhavenda ethnic group found in the northern region of South Africa. The Vhavenda group is an admirable example, as they possess a rich IK in their culture and traditional African religion and technology.

* Corresponding author.
E-mail addresses: 201101166@student.uj.ac.za (O.L. Malapane), nchanza@uj.ac.za (N. Chanza), wmusakwa@uj.ac.za (W. Musakwa).

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Studies on IK in the Vhavenda ethnic group have focused on the use of IK in ethnobotany, climate change adaptation using IK, and its use in natural resource management and agriculture (Constant, 2016; Kom et al., 2023; Castiano and Mkabela, 2010). Nevertheless, little focus has been put on the transmission of IK (Mistry, 2009; Magni, 2017). Thus, our study attempts to understand the extent to which local observations of IK transmissions are made through the lived experience of the Vhavenda people in the Vhembe district of South Africa. The study is important in documenting the importance of this knowledge transmission, preservation, and storage and its various benefits. The study was conducted in a remote area within the district where there is still visible evidence of IK practices. The focus was on demonstrating how the Vhavenda people transmit their traditional knowledge and how valuable IK is within the study area. It highlights what has been established by existing literature and reviews the shortcomings in the transmission and practices of IK, particularly in South Africa. This is followed by a description of the study area and the methods used for data collection. The results and discussion are then presented, followed by concluding remarks.

2. Related works

Traditional knowledge is similar to Western science based on observation but differs fundamentally from science (Berkes et al., 2000a). Traditional knowledge has become a major emerging research topic in the current epoch (Qwabe, 2020; Gómez-Baggethun, 2022; Morolo, 2004). With that said, the term traditional knowledge has various names that refer to it, such as rural knowledge, folk knowledge, local knowledge, indigenous technical knowledge, indigenous knowledge, and many more (Malapane et al., 2022; Mapara, 2009; Altieri, 1999). We have adopted the term Indigenous Knowledge (IK) for this study. Berkes (1992), however, has indicated that the term IK is ambiguous in its use and draws attention to questions concerning the cultural dynamics of such knowledge systems (Berkes and Folke, 1992). One can question what exactly IK is, but the various definitions of IK essentially denote the exact meaning of the term. Historical traditions can describe IK as a set of knowledge systems held by people outside Western science, encoded within rituals and everyday cultural practices unique and specific to a particular region and culture (Barth, 2002; Maweu, 2011; Madlela, 2017). Table 3 provides an overview of IK.

Table 1 shows that IK is a dynamic knowledge system that has a holistic outlook and is adaptive by nature (Knopf, 2015). It is gathered over generations by observations and experiences that are orally passed down from one generation to another so traditions can continue (Ohmagari and Berkes, 1997). Due to its oral transmission, IK is normally more transparent and willingly accessible to communities (Tharakan, 2017). In addition to agriculture, animal husbandry, child raising, education, medicine, and natural resource management, IK is used in many fields (Gope et al., 2017; Mazzocchi, 2018; Higgs and Van Niekerk, 2002). There has consequently been much discussion of how traditional knowledge can be analysed on several levels, according to some who describe it as a knowledge-practice-belief system (Berkes, 2012). There are four levels of analysis in IK and management systems, the first being local and empirical knowledge (Berkes, 2012). A resource management system, using local environmental knowledge and appropriate tools, techniques, and practices, is the second level. Third, management knowledge requires appropriate social institutions, rules-in-use, values, and relationships. The worldview is the final level of analysis, which determines how people view the environment (Berkes et al., 2000; Berkes, 2012).

Additionally, this knowledge system is used daily in various activities such as subsistence farming, food preparation, health care, decision-making, education, marital practices, risk management, and natural resource management (Zidny et al., 2020; Briggs, 2013). Not all IK systems are ecologically adaptive, and some of them have become unsustainable in the face of changing conditions over time (Berkes et al., 2000; Ahsan et al., 2021). For example, many indigenous groups collect firewood as part of their daily lives, thus contributing to deforestation, the loss of natural habitats, and ecosystem destruction (Kekana et al., 2023). Such activities are aspects of IK. They are considered social capital and sustain many rural livelihoods such as those of the Vhavenda ethnic group in the northern part of South Africa (Islam, 2013).

Table 1
Indigenous knowledge definitions, characteristics, and components.

Definitions	Characteristics	Components (Berkes, 2012: 19)
Warren (1991) defines IK as local knowledge uniquely central to a specific culture or society.	It is bound locally to a specific region belonging to the community.	Local knowledge (species-specific, identification, life histories, distributions)
Johnson (1992) defines IK as "a body of knowledge built up by a group of people through generations living in close contact with nature".	It is not documented, leading to it being lost.	Resource management systems (integrating local knowledge with practice)
IK is a set of knowledge unequally defined to a certain culture and society, and it forms the basis for local decision-making within local communities (World Bank, 1998).	IK is more transparent and is openly accessible to communities.	Social institutions (rules and codes of behaviour)
According to Masango (2010), IK is the entirety of all knowledge and practices rooted in experiences and observations of local people.	It is closely related to the survival and sustenance of many people worldwide.	World view (religion, ethics, and broadly defined belief systems).

Research into local Indigenous knowledge includes the following papers. Mutshinyalo and Siebert (2010) explored the role of traditional Vhavenda myths in conserving biodiversity and ecosystems, highlighting their significance in prohibiting unsustainable use of biodiversity. Mdhluli et al., (2013) focused on the need for intergenerational transmission of culturally unique knowledge for future generations through digital libraries. Masevhe et al. (2015) conducted an ethnobotanical survey on medicinal plants found in the Venda region that manage candidiasis and related fungal infections. Makgopa and Frangton (2016) inspected the perceived encounters posed by climate change on human-ecology interaction using an indigenous knowledge approach focusing on the Vhavenda people. Constant and Tshisikhawe (2018b) examined the Vhavenda people's role in indigenous plant use, practices, and beliefs, highlighting change drivers affecting knowledge attainment and transmission.

Nevertheless, the transmission of IK in indigenous communities has significantly disappeared due to the interference of foreign developments and concepts of Western technological innovation (Eyong, 2007; Lumpa, 2017). Extensive use of foreign practices, westernisation, and urbanisation has resulted in IK methods being rapidly eroded (Neitzel et al., 2019; Ferguson et al., 2017). Postcolonial society plays a vital role in the extinction of IK practices thus changing the shape and form of indigenous communities and their way of life (Trainor, 2020; Mbah and Fonchingong, 2019). The biggest downfall of IK is its oral transmission from one generation to another; disappearance is more prone to happen when the older generation passes away (Bruchac, 2014; Sraku-Lartey et al., 2017; Raphalalani, 2017; Schniter et al., 2018; Leal Filho et al., 2022).

The reality is that the younger generations are consequently culturally deficient, with no reference point, and need a clear sense of indigenous awareness, mindfulness, and self-identity (Kugara and Mokgoatšana, 2022). Documenting IK is therefore crucial for its

preservation and promotion against external factors like migration and modernization (Andrae-Marobela et al., 2012; Kaya and Seleti, 2013b). IK furthermore faces the challenge of being overlooked in policy and practice, despite its significant role in sustainable human life transformations (Vijayan et al., 2022). Despite global its recognition by IPBES, World Bank, and UNESCO, IK still lags in modern-day society. The South African government has implemented policies such as the Indigenous Knowledge Act of 2019 and the Intellectual Property System to recognise and protect traditional knowledge. However, these policy frameworks appear to be ineffective due to inadequate implementation, requiring further exploration and improvement.

Moreover, the disappearance of IK is directly linked to biodiversity loss, indicating the significant relationship between nature and culture (Convention on Biological Diversity, 2022). Such disappearance indicates a need to improve the intergenerational transmission of IK to supplement conventional education, preservation and documentation of IK through media (United Nations, 2021; Owiny et al., 2014). However, certain restrictions should be implemented to protect indigenous people and their knowledge (Deer, 2019). Proper storage and management of IK must be ensured, together with knowledge safety, to allow for reputable knowledge sharing, analysis, and manipulation by various professionals (Tripathi and Bhattarya, 2004).

Mdhluli et al. (2021) highlight that Africa's IK is on the brink of extinction due to outdated knowledge management methods, with indigenous people concerned about Western misappropriation. Nevertheless, African IK is crucial for the survival of communities, as most food is produced using traditional farming practices (United Nations Conference on Trade and Development, 2004). However, urbanisation, technological advancement, and Western lifestyles are posing a risk to these practices and knowledge systems (Stephens et al., 2006). IK in Africa sustained local and indigenous communities for generations before colonialism (Mapira and Mazambara, 2013). Such colonisation weakened the significance of Indigenous Knowledge (IK) while promoting Western knowledge systems (Kaya and Seleti, 2013). This has led to a dependence on Western knowledge systems, and Africa has long neglected its rich IK (Khumalo and Baloyi, 2017; Adeyeye and Mason, 2020). Moreover, African IK is threatened by a lack of adaptability in knowledge sharing, storage, and management, which is outdated with modern knowledge dissemination methods (Mdhluli et al., 2021). Consequently, various organizations and governments have made significant efforts to safeguard indigenous communities and their knowledge (Tsekea, 2016). These efforts yield results, but others are challenging due to the inherent nature of the knowledge system (Langill, 1999). Similarly, Other religions and modern education have significantly impacted the Vhavenda tradition and culture, resulting in the extinction of their way of life (Mdhluli et al., 2021).

3. Methods and materials

3.1. Geographical location

Located in the northern part of the Limpopo province of South Africa, the Vhembe district boundaries are shared by the Capricorn and Mopane District Municipalities in the east and west, respectively (Limpopo Department of Economic Development, 2019). Through the Kruger National Park, it shares its northern border with Zimbabwe, Botswana, and Mozambique in the southeast (Vhembe District Municipality, 2022). The name Vhembe comes from the Venda language and means Limpopo River, and the district is part of the Great Limpopo Transfrontier Park (Department of Cooperative Government and Traditional Affairs, 2020). The landscape in Vhembe is beautifully filled with a rich history connected through forests, waterfalls, lakes, and deep valleys deeply rooted in culture (Limpopo Department of Economic Development, 2019). The district boasts a rich cultural and traditional diversity rooted in spiritual significance, diverse ethnobotanical systems, and a distinctive way of life. The study area was chosen due to its rich indigenous practices, beliefs, culture, traditions, knowledge of plants, medical practices, water management, farming experiences, animal health, and behaviour. The area is inhabited by locals whose extended history provides valuable experiences of IK of the Vhavenda people.

The data were collected within Masisi and surrounding villages (Duluthulwa, Tshikuyu, Dovho, Mutele, and Sanari) as illustrated in Fig. 1. The villages between Thohoyandou and Musina are characterised by a flat landscape, low rainfall, and drought, averaging 300–400 mm annual rainfall. (Shikwambana et al., 2021). The study area was situated near Kruger National Park and Pafuri Nature Reserve and is governed by traditional councils and chiefs known as Bamusanda. The Mutele Traditional Authority makes decisions in the Masisi area. The road leading to Pafuri Gate is popular among tourists visiting Kruger National Park, and the area is agricultural, with farming and livestock practices.

3.2. Data collection and analysis methods

The study utilised a mixed methods approach to comprehensively understand indigenous transmission methods and practices. The researcher utilised questionnaires and interviews to conduct in-depth research on a dynamic topic, obtaining detailed insights from respondents and revealing their experiences. This approach allowed exploring the transmission of IK through respondents' experiences and their interconnectedness with action, history, culture, and social connections.

Purposive sampling was used to select participants with in-depth experience and understanding of IK, based on their long history and ages. The qualitative approach enabled the gathering of data from community residents aged 50 and above, using questionnaires and interviews. The sample size was 136, with interviews with traditional leaders, entrepreneurs, farmers, and healers. Since the population was situated in a traditional tribal area ruled by traditional councils, the researcher contacted one of the traditional council members to assist in the data collection process. The traditional council member agreed to a formal meeting with the two traditional council authorities for the researcher to present the study's purpose and ask permission to conduct the study. Such traditional areas in certain regions require proper permission from chiefs and local authorities to enter, as it is customary and part of their cultural traditions.

The traditional Mutele and Tshikondamalema councils gave permission to collect data in Masisi and surrounding villages. The Mutele and Tshikondamalema traditional council consists of traditional leaders, chiefs (Musanda), and headsmen around the Masisi area. A key traditional council member also assisted the researcher in sourcing suitable field assistants, and the university criteria for participation were shared with this member. Two university graduates who had prior data collection experience were responsible for translating questionnaires and interviews into the local Tshivenda language and turning the responses into English. The field assistants were trained to guide them on the research, ethical protocols, and COVID-19 regulations.

The data were analysed using Excel spreadsheets, recorded, coded, and sorted into qualitative and quantitative data. MATLAB and NVivo were used for descriptive and thematic analysis, identifying themes, patterns, and statements for interpretation. The University of Johannesburg's Faculty Ethics and Plagiarism Committee approved the research, ensuring that required ethical standards and obtaining consent from participants through questionnaires were met. All Covid-19 regulations, such as wearing a mask, sanitising, and maintaining a 1.5 m social distance, were strictly enforced. The study's purpose was clearly explained to participants, who were informed about privacy, confidentiality, and their rights, with informed consent obtained and the ability to withdraw at any time.

4. Socio-economic and demographic characteristics

The table (Table 2) below provides a comprehensive overview of the

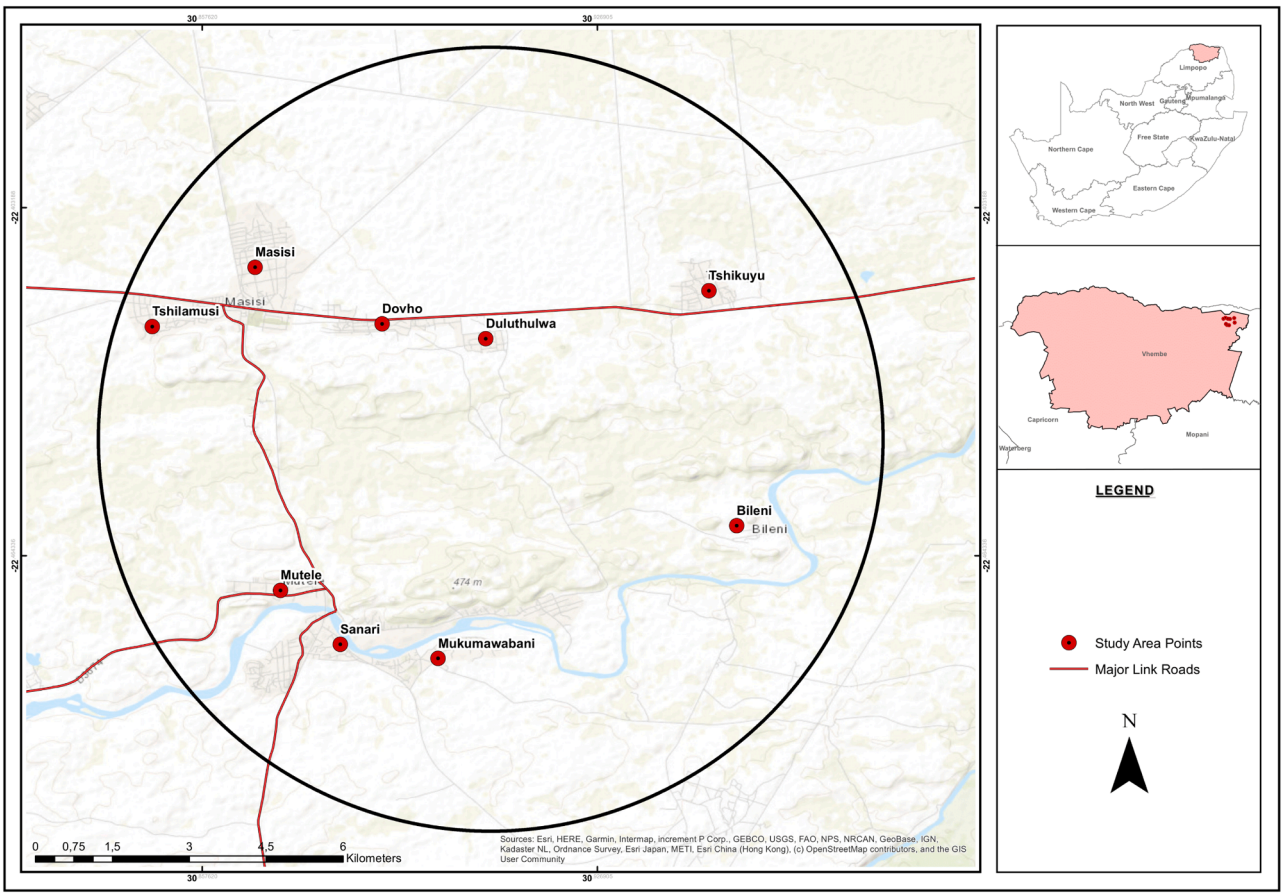

Fig. 1. Map of sites within the study area where data were collected.

Table 2
Socio-demographic characteristics.

Gender	%	Age	%	Educational Level	%	Employment Status	%	Income Level	%
Male	40	50–54	33	Never attended school	25	Unemployed	67	<R 3000	89
		55–59	20	Primary level	33	Employed	42	R 3001 – R5000	5
		60–64	44	Secondary level	36	Pensioner	20	R10 001 – R 15 000	4
		70–74	4	Tertiary level	5	Self Employed	9	R15 001 – R 20 000	2
Gender	%	Age	%	Educational Level	%	Employment Status	%	Income Level	%
Females	60	50–54	19	Never attended school	36	Unemployed	54	<R 3000	89
		55–59	5	Primary level	31	Employed	6	R 3001 – R5000	6
		60–64	58	Secondary level	22	Pensioner	28	R10 001 – R 15 000	4
		70–74	7	Tertiary level	11	Self Employed	11	R15 001 – R 20 000	1
		80+	1						

study area's socio-demographic and economic attributes. The dominant age group of participants was 60–64, followed by 50–54, 55–59, 70–74, and 80. Sixty of the study participants were females. In the study area, unemployment was notably high, particularly among females, with over 60 % of women and roughly 54 % of men reporting unemployment. The number of educated men in the study region was higher than that of women, with 36 % of males having secondary education compared to 22 % of females. Interestingly, more females (11 %) have tertiary education compared to only 5 % of males. Regarding household income levels, most participants indicated having a household income below or equal to R3000 per month, followed by R3001 – R5 000 per month. Hence, most households in the study area live below the poverty line and do not meet the standard household income as per South African labour law.

Fig. 2 depicts the various types of livelihoods present in the study area. Farmers, fuel wood collectors, traders, herbalists, traditional healers, wood carvers, and hunters are among the most common

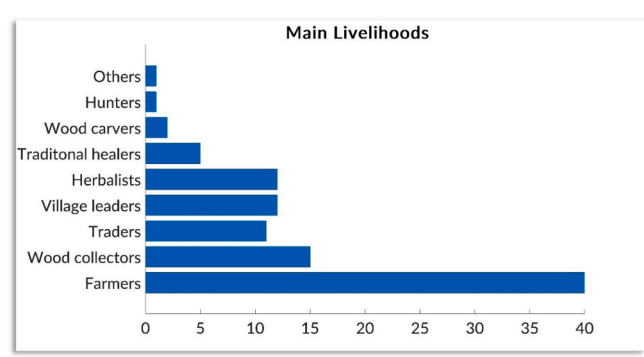

Fig. 2. Categorisation of various livelihoods found in the study area.

livelihoods. These livelihoods are consistent with household income levels in the study area, as shown in Table 2. The transmission of IKS in the study area is linked to various livelihoods as shown in Fig. 2. These include traditional customs, principles, norms, standards, taboos, poems, proverbs, practices, and teachings, all of which contribute to the community's rich indigenous knowledge and culture. The preservation of IKS is critical for preventing its extinction. As a result, the government must devise effective policies to safeguard and advance indigenous knowledge systems. These policies should be developed in collaboration with a variety of stakeholders, including local non-governmental organisations and indigenous communities. By preserving and transmitting IKS, we can effectively address global issues like climate change, biodiversity loss, agricultural sustainability, and ecosystem stability. Thus, incorporating IKS into mainstream practices is critical.

5. Results and discussion

5.1. Transmission of indigenous knowledge systems in the vhavenda community

Participants were questioned about the various methods of transmitting IKS in the study area. It was found that folklore (49 %) was the most used method of IKS transmission in the area (Fig. 3). Folklore in this context refers to traditional tales, legends, proverbs, poems, myths, taboos, and jokes that are passed down orally. Informal education emerged as the second most frequent form of IKS transmission, with participants indicating that IKS is transmitted through observation, participation, and oral literature. Additionally, traditional songs (7 %) were identified as the third most common form of IKS transmission. These songs are essential to the culture and religious practices, depicting the ancestral history and philosophical issues. Lastly, less frequent forms of transmitting IKS included Television (5 %), Radio (4 %), Spirit mediums (2 %), formal education (1 %), and the internet (1 %).

The research study revealed that the Vhavenda Indigenous Knowledge System (IKS) is predominantly passed down through oral traditions such as tales, legends, myths, proverbs, poems, and jokes, which are integral to the Vhavenda cultural identity as per Fig. 3. Participants emphasised that their customs and traditions are transmitted through proverbs, songs, rituals, and practices taught by elders. For instance, one of the participants stated: *"As children, we would gather around the fire at night to listen to the captivating tale of King Singo and his enchanted drum, as narrated by our elder. According to Venda folklore, the sound of the drum would instil fear in the enemy's hearts, causing them to flee. Some believe King Singo vanished from his kraal one night, taking the unique drum with him, and was never seen again"*. The narratives of these stories, myths, and legends are conveyed through oral traditions carried across generations. This transmission mode poses the risk of potential inaccuracies and difficulties in ensuring the verifiability and authenticity of the original versions. Therefore, the community around the study area have established a showcase museum at the Awelani Lodge to educate others about their traditional way of life in the Masisi area and to preserve their Vhavenda IKS. This demonstrates the Vhavenda's commitment to preserving and perpetuating their knowledge system and highlights their significance.

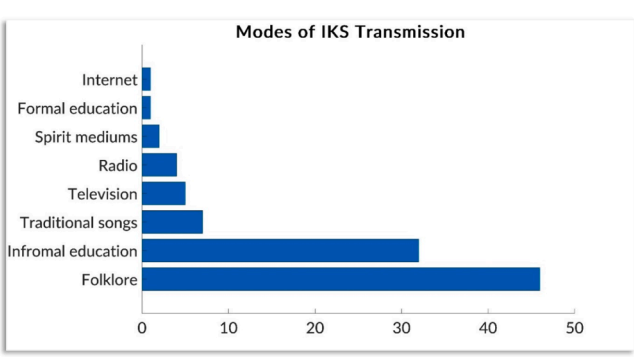

Fig. 3. Various Modes of how IKS is transmitted in the study area.

IKS in the region is also transmitted through informal education, which conveys knowledge through observations, participation, and oral literature. For instance, in the Vhavenda community, informal education occurs through practices such as farming, enabling the transfer of knowledge from the older to the younger generation through observations, practices, and experiences. Additionally, IKS is transmitted through the traditional authority councils of the Vhavenda community, such as the Mutele and Tshikondamalema councils. These convene to discuss various community matters and offer solutions that adhere to cultural standards and norms to address the community's daily challenges. One participant explained, *"The traditional councils serve as the custodians of the Vhavenda people. We gather to deliberate on issues about the community's well-being. Our guidance is grounded in customary laws and principles passed down by our forefathers, aiming to provide solutions and lead the community"*. This highlights that IKS transmission is not solely about knowledge but also embodies a traditional way of life which is intangible.

Another method of transmitting IKS in the region is through traditional songs, although this mode is not as prevalent as other methods. These traditional songs are part of the region's folklore and convey historical events, cultural identity, traditional philosophies, myths, legends, and taboos. They hold significant importance in the Vhavenda culture and history.

Traditional songs are another way to transmit IKS in the region, though they are less common than other methods. These traditional songs are part of the region's folklore and represent historical events, cultural identity, traditional philosophies, myths, legends, and taboos. They play a significant role in Vhavenda's culture and history. One participant expressed, *"Our traditional knowledge is conveyed through traditional songs, which we have learned carry important messages through listening, participation, imitation, and repetition."* Another participant emphasised, *"These traditional songs encompass legendary tales, cultural poems, proverbs, and family lineages of our forefathers, which we are obligated to pass down to the next generation."* All of the mentioned methods of IKS transmission in the Vhavenda community are similar and interconnected, as they all involve oral communication and knowledge transfer through listening, imitation, repetition, participation, demonstration, and observation.

Television, radio, spirit mediums, formal education, and the Internet were among the less popular channels for transmitting IKS in the region. This is most likely due to the area's rural location, where technological advancements are less common and educational institutions may be scarce due to slow development and transition. Spiritual mediums' practices are often shrouded in secrecy and mystery. These people are highly respected in their community because they are thought to be able to communicate with ancestors and other spiritual entities, as well as predict the future (Kaya, 2016). As a result, their knowledge is closely guarded and only shared with those who have similar gifts or follow the same traditions.

5.2. Forms of indigenous knowledge systems transmitted, valued and practised in the vhavenda community

Fig. 4 depicts the three main categories of IKS: cultural traditions, connection to a place, and environmental ethics. In the study area, cultural traditions emerged as the most important category of IKS across all three assessed categories, surpassing both environmental ethics and a sense of place.

The three categories of IKS abovementioned in Fig. 4 are defined as follows: cultural traditions are a set of beliefs, principles, norms, standards, or ways of life that a particular society or group has followed and practised for a long time and is still cohesive and intact (Valadez, 2012;

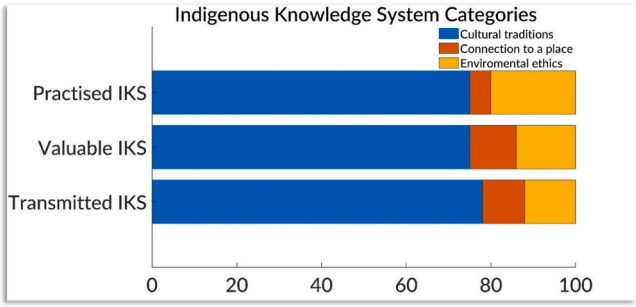

Fig. 4. Categories of indigenous knowledge transmitted, most valuable, and practised in the study area.

Baker, 2013). Connection to a place is the affective bond between people and their surroundings which can be described as a strong sense of place, frequently combined with a sense of cultural identity, emotional and spiritual value, and a sense of belonging and rootedness (Pritchard, 2023). Environmental ethics refers to values, principles, societal attitudes, actions, and policies a particular community or society upholds to protect and sustain biodiversity and ecological systems (Minteer, 2014).

For the study cultural traditions include a community's long-standing beliefs, principles, norms, and standards. For example, a participant indicated that: "*it is customary for young people to greet elders whenever they meet them. Also, it is expected that if an elder boards a bus or taxi and no seats are available, a young person should offer their seat to the elderly*". Another participant emphasised the prominent tradition of circumcision, which is performed as part of the transition into adulthood: "*When males and females reach the age of majority, they must be taken to the initiation school to be circumcised. Here we learn about how one should behave.*" The connection to a place represents a profound emotional attachment between a community and its surroundings, reflecting a strong sense of place, belonging, and culture. Participant, for instance, stated that: "*water is an important element in our culture; water spirits (Zwidutwane) live in sacred areas such as Lake Fundudzi, and we take food as an offering to honour them.*" Environmental ethics is a community's moral obligation to protect the environment through established norms, standards, myths, taboos, and principles rooted in tradition and culture. The Vhavenda people's environmental ethics centre on protecting forests, mountains, hills, and caves because these are where their ancestors lived. For example, participants highlighted that: *"the Thathe Vondo forest is believed to be guarded by a white lion that protects the ancestors of the Netshidzivhe clan, and entry into such sites is forbidden"*.

Fig. 4 indicates that cultural traditions are the most valued, practised, and transmitted in the study area. This demonstrates the Vhavenda people's strong connection to their dynamic cultural traditions. For example, one of the participants mentioned that *"we perform rituals to honour and appease the ancestors for favourable weather and a productive harvest or seeking ancestral blessings before embarking on long journeys"*. Environmental ethics emerged as the region's second most important category of transmitted, valued, and practised IKS. This emphasises the Vhavenda people's strong moral obligations to their environment, as evidenced by practices such as not hunting pregnant animals and preserving the traditional significance of the Muvhombew tree. According to Muralikrishna and Manicka (2017), the cultural environment is inextricably linked to taboos and myths, reflecting the community's profound and esoteric relationship with nature. These findings imply that the Vhavena community maintains a harmonious relationship with their natural environment through ethics rooted in traditional ecological knowledge systems.

Lastly, connection to a place was ranked third in terms of IKS transmission, value, and practice in the study area. This type of connection represents cultural identity, sense of belonging, and attachment to the environment or a specific location in the region. This attachment could be due to spiritual, historical, or cultural significance. It extends beyond physical attachment to reflect intangible values. For example, one participant mentioned that certain areas, such as the Mashovhela Rock Pool and Lake Fundudzi, are spiritually and historically significant in Vhavenda culture. These sites are important to their traditional identity and are revered within the community. To support this, Pikirayi and Magoma have stated that sacred landscapes in the Vhavenda community are associated with the community's self-identity, well-being, and sense of place. Legends, taboos, and myths serve as reflections of these areas (Pikirayi and Magoma, 2021).

5.3. *Drivers of indigenous knowledge system disappearance in the study area*

Fig. 5 below illustrates the primary threats leading to the disappearance of Indigenous Knowledge Systems (IKS) within the Vhavenda community. Participants were asked to rank five drivers based on their prevalence using a scale ranging from "higher" to "none". The results revealed that the invasion of technology, the failure of elders to pass on knowledge, and the forgetting of IKS were identified as the most prevalent drivers of IKS disappearance. At the same time, extinction was considered the least prevalent. According to the study, the drivers identified as less effective and forgotten were found to have a moderate impact on the disappearance of Indigenous Knowledge Systems (IKS) in the study area. This suggests that these specific drivers may have contributed to the decline of IKS within the region.

Table 3 below provides the meaning behind each term used on the y-axis of Fig. 5 above to provide a comprehensive understanding of the meaning behind each term used as per the research conducted in the study area.

According to the results in Fig. 5, the decline of IKS in the region is primarily due to community members' widespread adoption of technology. This simply means that the spread of technology in indigenous communities affects native culture (Zulfadrim et al., 2021). As a result, the community's ability to practise its own culture is limited, which has a negative impact. The second and third most common reasons for IKS disappearance were elders' reluctance to pass on knowledge and community members' forgetfulness of the knowledge. This was attributed to a reliance on modern practices and technology, as well as a lack of interest on the part of the younger generation in learning indigenous skills and knowledge.

What is interesting about the findings is that extinction was found to be a non-driver of IKS disappearance, which is because there are still traces of indigenous knowledge practises in the study area, and despite modernity's impact, the Vhavenda people are still recognised as the most culturally inclined people. Moreover, the reduced effectiveness of the knowledge was found to have a moderate impact on the local IKS. For example, one participant stated: "*Some of our agricultural methods, such as using cattle or donkeys to harvest grass, are not as effective as they once were and this type of method is time-consuming, and there are more advanced methods for harvesting, such as using tractors.*" All of the

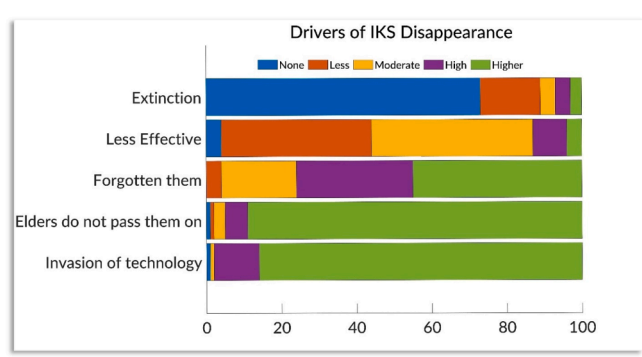

Fig. 5. Disappearance Drivers of Indigenous knowledge in the study area.

Table 3
Terminology and meaning of each phrase used on the y-axis of Fig. 5.

Terminology	Meaning in terms of study
Extinction	The dying out, loss or disappearance of local or traditional knowledge.
Less Effective	Local or traditional knowledge lacks the ability or form to perform adequately.
Forgotten them	the inability to remember or recall local or traditional knowledge.
Elders do not pass them on	Lack of willingness or desire of elders to transfer local or traditional knowledge to the youth.
Invasion of technology	The growing influence and penetration of technology in indigenous communities leading to their inability to practice local culture and rely more on western knowledge.

aforementioned drivers pose a significant threat to the preservation of traditional knowledge systems, limiting the community's ability to maintain cultural practices and apply this knowledge in daily life. This harms these communities because it erodes their cultural identity and undervalues their knowledge systems, traditional way of life, and philosophical beliefs. Zulfadrim et al. (2021) investigate the impact of technology on culture within indigenous communities and provide evidence for the current findings.

Furthermore, the failure to transmit knowledge systems within indigenous communities can cause these valuable knowledge systems to deteriorate and, ultimately, disappear. This occurs when elders fail to pass on their wisdom to the next generation, leading to community members forgetting this knowledge and failing to evolve and adapt. This, in turn, leads to cultural changes, the adoption of modernised lifestyles, the loss of traditional territories, environmental consequences, and cultural shifts. For example, the Vhavenda people face the risk of losing their indigenous knowledge, culture, and traditions due to a lack of generational transmission. One participant stated that elders are hesitant to pass on cultural knowledge and wisdom to the next generation due to a lack of interest, participation, and respect for culture and tradition. This disrupts knowledge transfer, longevity, and preservation, resulting in cultural identity crises for future generations. Even if knowledge does not completely vanish, the essence of the culture gradually fades.

6. Study limitations

Considering the research's nature as a case study, it's important to acknowledge certain limitations. One such limitation pertains to the potential transferability of the findings to other areas, however, it's crucial to note that the results remain pertinent and impactful. Additionally, time constraints posed another possible limitation, as the significant expanse of the region prevented us from collecting data from all areas within it. The research data were gathered during the COVID-19 pandemic when there were restrictions on face-to-face interactions. Nevertheless, the researcher complied with all regulations, including maintaining a safe distance between individuals, always wearing a mask, and regularly sanitising hands.

7. Conclusion

The research contributes to the body of knowledge by demonstrating the importance of transmitting Indigenous Knowledge Systems (IKS) for indigenous and local communities, as well as ensuring the sustainability of natural resources. Additionally, the research identifies the various factors that impact this type of knowledge within the study area and how they affect indigenous cultures and traditions. The research indicates that IKS is fundamentally important to indigenous identity, culture, languages, heritage, and livelihoods. The transfer of IKS from one generation to the next must be preserved and promoted, as these interactions represent the knowledge of the local ecologies. Through the results, it can be concluded that IKS constitute a fundamental component of the cultural and traditional identity within the Vhavenda community. The erosion of these systems presents significant threats to the socio-cultural dynamics and the preservation of identity in the region. The urgency of safeguarding and facilitating the intergenerational transmission of this knowledge has highlighted the necessity for policy interventions to bolster such systems' resilience. This necessitates a concerted effort involving indigenous populations, governmental stakeholders, Non-Governmental Organizations (NGOs), and international bodies advocating for indigenous communities' rights and knowledge preservation. The encroachment upon Indigenous Knowledge Systems by technological advancements, the discontinuity of knowledge passage across generations, and the diminishing applicability of traditional knowledge underscores a critical challenge—these systems' adaptability and evolution are at risk. Indigenous communities moreover contribute to this threat due to neglect in valuing and transmitting their knowledge; further imperilled by an apprehension towards the relevance of their IKS in the face of modernity. Nevertheless, within the Vhavenda community, the persistence of IKS, encapsulating cultural traditions, environmental stewardship and a profound connection to their lands remains the cultural backbone of the region. Thus, preservation endeavours within the community primarily manifest through traditional authorities, national heritage sites, and cultural museums, where IKS intertwines with daily livelihoods and the transference of knowledge persists. IKS transmission within the Vhavenda predominantly occurs orally, through folklore, informal education methods such as farming practices, and traditional songs encapsulating stories, myths, legends, and taboos. Despite the pervasive threat posed by technological invasion, current modes of knowledge dissemination, such as television, radio, and the internet, remain underutilised for transmitting local knowledge. This constitutes a critical disadvantage for preserving IKS as these digital platforms offer unparalleled longevity and breadth in knowledge dissemination. Therefore, an imperative exists for integrating Indigenous Knowledge Systems with contemporary knowledge frameworks. Such integration is pivotal in devising sustainable solutions to global challenges, including climate change, ecosystem degradation, environmental decline, and the erosion of cultural identity and heritage in indigenous communities. Future research can explore the multifaceted implications of policy and practice on indigenous communities, emphasising expansive geographical scopes and delving into specific IKS themes, such as the roles of smallholder farmers and women in perpetuating indigenous knowledge.

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CRediT authorship contribution statement

Walter Musakwa: Validation, Supervision, Software, Project administration, Data curation. **Olgah Lerato Malapane:** Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Nelson Chanza:** Writing – review & editing, Validation, Supervision.

Declaration of Competing Interest

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Data availability

I have attached the supporting document for the questionnaire and interview schedule.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.envsci.2024.103861.

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